

Project Design Phase-II Data Flow Diagram & User Stories

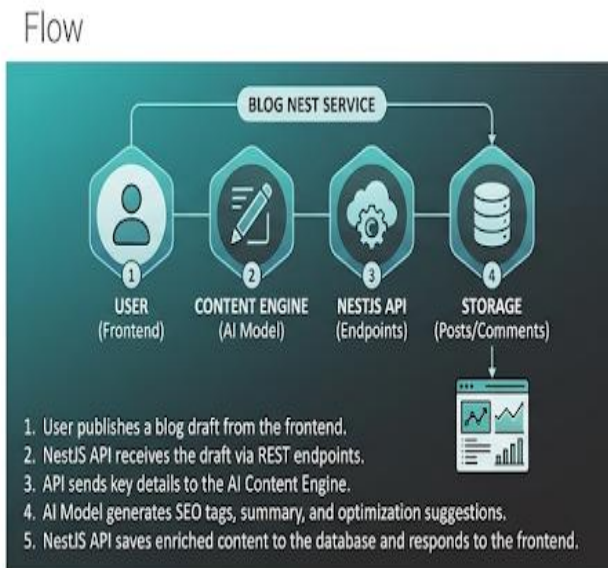
Date	28 SEPTEMBER 2026
Team ID	SWTID-2026-4636
Project Name	AI BLOGNEST API
Maximum Marks	4 Marks

Data Flow Diagrams:

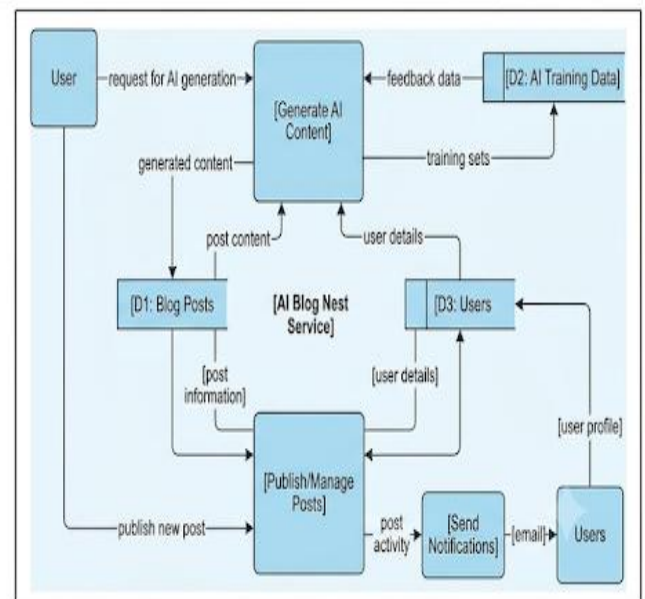
A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

store.

Example: (Simplified)

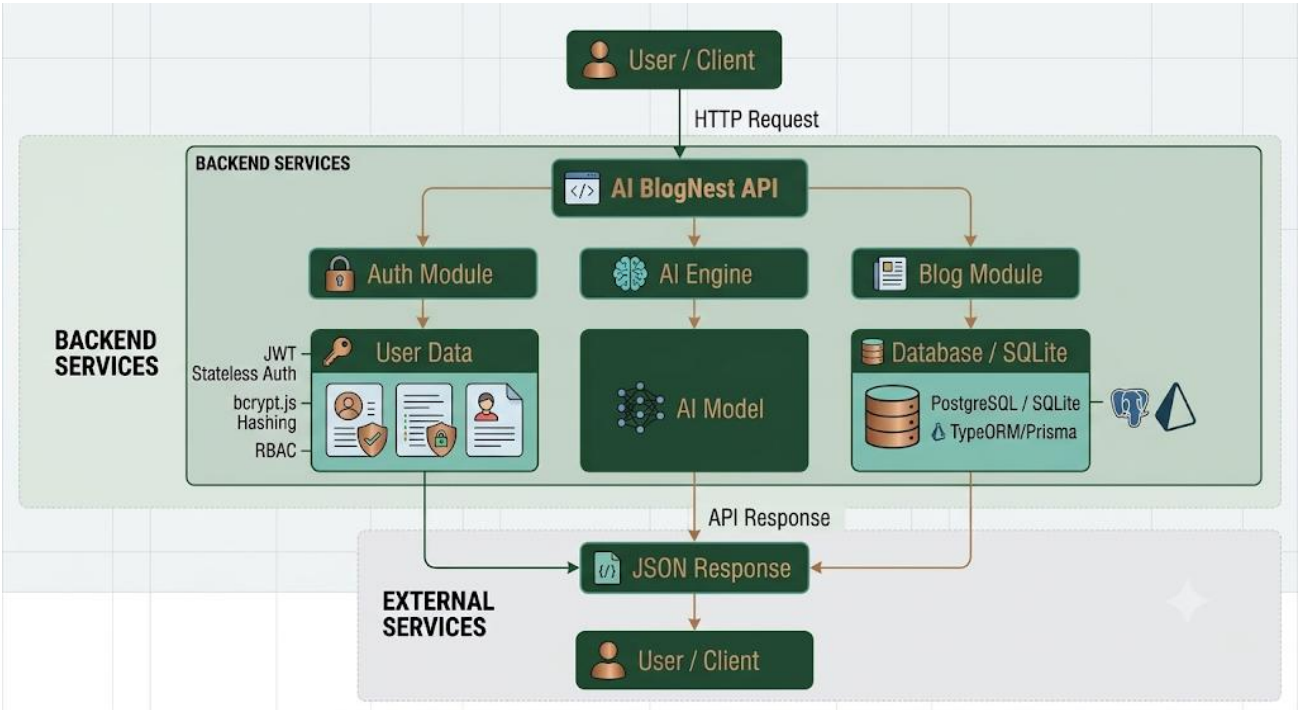


Example: DFD Level 0 (Industry Standard)



Data Flow Diagram & User Stories

- AI BlogNest API - Basic Data Flow



User Stories - AI BlogNest API

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Customer (Web User)	User Registration	USN-1	As a user, I can register for AI BlogNest by entering my name, email and password.	User account is created successfully when valid details are provided.	High	Sprint-1

Customer (Web User)	User Registration	USN-2	As a user, I should receive an appropriate response after successful registration.	Registration success message is returned and user details are stored securely.	High	Sprint-1
Customer (Web User)	User Login	USN-3	As a user, I can log into AI BlogNest using my registered email and password.	Valid credentials allow login and return an authentication token.	High	Sprint-1
Customer (Web User)	Authentication	USN-4	As a user, I can access protected APIs after successful authentication.	Valid token allows access to protected endpoints.	High	Sprint-1
Customer (Web User)	User Profile	USN-5	As a user, I can view my profile after logging in.	Authenticated user details are returned correctly.	Medium	Sprint-1
Customer (Web User)	AI Blog Generation	USN-6	As a user, I can generate blog content by entering a topic and category.	AI returns relevant blog content for the provided topic and category.	High	Sprint-2
Customer (Web User)	AI Blog Generation	USN-7	As a user, I can generate suitable blog title suggestions using AI.	API returns one or more relevant title suggestions.	Medium	Sprint-2

Custome r (Web User)	AI Blog Summariza tion	USN-8	As a user, I can summarize existing blog content using AI.	API returns a concise summary of the submitted content.	High	Sprint- 2
Custome r (Web User)	Input Validation	USN-9	As a user, I should receive a validation message when required information is missing.	API returns an appropriate validation error for empty or invalid input.	High	Sprint- 2
Custome r (Web User)	Blog Manageme nt	USN-10	As a user, I can save generated blog content.	Valid blog details are successfully stored in the database.	High	Sprint- 3
Custome r (Web User)	Blog Manageme nt	USN-11	As a user, I can view all my saved blogs.	API returns the user's available blogs.	High	Sprint- 3
Custome r (Web User)	Blog Manageme nt	USN-12	As a user, I can view a specific blog using its ID.	Correct blog details are returned for a valid blog ID.	Medium	Sprint- 3
Custome r (Web User)	Blog Manageme nt	USN-13	As a user, I can update an existing blog.	Updated blog content is stored successfully .	High	Sprint- 3

Customer (Web User)	Blog Management	USN-14	As a user, I can delete an existing blog.	Selected blog is removed successfully from the database.	High	Sprint-3
Customer (Web User)	Blog Search	USN-15	As a user, I can search for blogs using relevant keywords.	API returns blogs matching the search criteria.	Medium	Sprint-4
Developer / Administrator	API Validation	USN-16	As an administrator, I can validate API requests and handle errors.	Invalid requests return suitable HTTP status codes and error messages.	High	Sprint-4
Developer / Administrator	API Testing	USN-17	As a developer, I can test all AI BlogNest APIs using Postman.	Authentication, AI and blog endpoints return expected responses.	High	Sprint-4
Developer / Administrator	API Documentation	USN-18	As a developer, I can document API endpoints, parameters and responses.	API documentation contains endpoint details and request/response examples.	Medium	Sprint-4
Blogger / Content Creator	AI Content Generation	USN-9	As a blogger, I can enter a topic and generate blog content using AI.	Relevant content is generated based on the provided topic.	High	Sprint-3

Blogger / Content Creator	AI Content Generation	USN-10	As a blogger, I can modify AI-generated content before publishing it.	Generated content can be edited and saved.	Medium	Sprint-3
Blogger / Content Creator	Blog Organization	USN-11	As a blogger, I can categorize my blog posts.	Blog posts can be assigned to appropriate categories.	Medium	Sprint-3
Blogger / Content Creator	Search	USN-12	As a user, I can search for blogs using keywords or titles.	Matching blog posts are displayed in the search results.	Medium	Sprint-3
Blogger / Content Creator	Security	USN-13	As a user, I can access my blog data securely.	Unauthorized users cannot access protected blog data.	High	Sprint-4
Blogger / Content Creator	Error Handling	USN-14	As a user, I can receive clear messages when invalid information is submitted.	Appropriate validation and error messages are displayed.	Medium	Sprint-4
Administrator	User Management	USN-15	As an administrator, I can manage registered user accounts.	Administrator can view and manage authorized user information.	High	Sprint-4

Administrator	Blog Management	USN-16	As an administrator, I can monitor blog content stored in the system.	Administrator can view available blog records.	Medium	Sprint-4
Administrator	Database Management	USN-17	As an administrator, I can maintain user and blog information securely.	Data is stored and retrieved correctly from the database.	High	Sprint-4
Developer / Tester	API Testing	USN-18	As a developer, I can test the BlogNest REST APIs using Postman.	All major API endpoints return the expected responses.	High	Sprint-4
Developer / Tester	System Testing	USN-19	As a tester, I can test both valid and invalid user scenarios.	Positive and negative test cases are completed successfully.	High	Sprint-4
Project Team	Deployment	USN-20	As a project team, we can prepare the AI BlogNest API for final deployment.	Application is tested and ready for deployment.	High	Sprint-4